From Continual Learning to Self-Evolving Agents

Tongtong Wu (Monash University) Dong Gong (UNSW Sydney) Reza Haffari (Monash University)

Abstract. Continual learning has focused on acquiring new knowledge without catastrophically forgetting prior capabilities. Modern AI systems face a broader challenge: data, knowledge, user needs, tools, environments, and system architectures all evolve over time. This tutorial presents continual learning as the foundation for building self-evolving agents. We organise the material around four questions: what to learn, when to adapt, where to store new knowledge, and how to verify an update. We revisit continual learning for foundation models, develop an adaptation hierarchy from retrieval and external memory through prompts, adapters, and modular experts to backbone updates, and treat memory as an active decision mechanism. We then describe agents that learn from their own operation, and a closed loop in which they diagnose capability gaps, adapt selectively, verify improvements, and consolidate. The central message is that effective self-evolution requires selective, localised, and verifiable adaptation rather than unrestricted self-training.

1. Goals and Learning Outcomes

The tutorial targets ML researchers and practitioners familiar with deep learning and foundation models; no background in continual learning or agents is assumed. Our goal is to replace a catalogue of methods with a small set of transferable design decisions, so attendees will be able to:

- distinguish transient context shifts from persistent capability changes, and place any adaptation method within the four organising questions;
- select where an update should live, from context and retrieval to external memory, adapters, modular experts, or backbone parameters, given the persistence, scope, and timescale of the change;
- design a memory policy governing what experience is kept, abstracted, retrieved, and forgotten, and diagnose whether a failure calls for changing memory or parameters; and
- instantiate the self-evolution loop of diagnosis, adaptation, verification, and consolidation, and evaluate such systems beyond final-task accuracy.

2. Description and Outline

The tutorial is organised as six self-contained modules sharing a common spine, an adaptation hierarchy and a self-evolution loop, so each can be followed alone while the sequence builds one argument.

Module 1: Overview, From Continual Learning to Self-Evolving Agents (Wu, Gong, Haffari; 15 min). We map the vocabulary of continual learning, self-evolution, and LLM agents, distinguish transient context shifts from persistent capability changes, explain why repeated fine-tuning is inefficient, and introduce the four organising questions: what to learn, when to adapt, where to store, and how to verify [1,2].

Module 2: Continual Learning Foundations and the Foundation-Model Era (Gong, 25 min). We revisit replay, regularisation, parameter isolation, and expansion through their assumptions and stability–plasticity trade-offs [3], then examine how foundation models reshape forgetting and transfer, motivating selective rather than uniform updates [4]. We close with an adaptation hierarchy, context and retrieval, external memory, prompts and adapters, modular experts, backbone parameters, and a rule selecting where to update from the persistence, scope, and timescale of the change [5–7].

Module 3: Memory in Continual Learning and Self-Evolution (Gong, 30 min). We treat memory as an active decision mechanism rather than a passive store: experience selection, abstraction, retrieval, and deliberate forgetting [7,8]. We compare episodic and semantic, parametric and external, and passive and diagnosis-driven memory, and show why larger memory does not imply stronger long-term capability.

Module 4: Agents That Learn: Anatomy and Where Change Enters (Wu, 20 min). We describe the components of an agent that learns from its own operation, policy, context, memory, tools, skills, environment, and multi-agent topology, and the loop connecting them. Using measured evidence of deployment drift in tools, library versions, and knowledge sources [9], we map the adaptation hierarchy onto these components, separating component- from policy-level change.

Module 5: Self-Evolving Agents (Wu, 30 min). We develop a closed loop of experience, capability diagnosis, targeted assistance, selective adaptation, verification, and consolidation [10], organising methods into component-level evolution of memory, skills, and tools; policy-level evolution from trajectories and active interaction; test-time adaptation; and structural evolution of agent topologies [11].

Module 6: Evaluation, Open Challenges, and Conclusion (Haffari, 20 min). We cover introspection and self-evaluation as the gate on the loop [12], then evaluation beyond final-task accuracy: retention, adaptation speed, transfer, efficiency, robustness, and error accumulation. We close on open challenges, model–memory coordination, resource growth, reliable self-generated learning, and five principles: adapt selectively, localise stability–plasticity trade-offs, separate transient information from reusable capabilities, treat memory as active, and verify before consolidating.

3. Presenters

The 150 minutes are divided as follows: Module 1 jointly by all three presenters (15 min); Dong Gong, Modules 2–3 (55 min); Tongtong Wu, Modules 4–5 (50 min); Reza Haffari, Module 6 (20 min); closing discussion (10 min). All presenters will attend in person.

Tongtong Wu is a Research Fellow in the Department of Data Science & AI at Monash University (`Tongtong.Wu@monash.edu`). His research focuses on continual learning [1], adaptive language models and memory [9], and self-evolving agents [11].

Dong Gong is a Senior Lecturer in the School of Computer Science and Engineering at UNSW Sydney (`dong.gong@unsw.edu.au`), where he leads the Artificer AI Lab. His work targets reliable, self-improving AI for open-ended, continuously changing environments: continual learning [5], memory mechanisms and dynamically expandable architectures (dynamic MoE) [6,7], and post-training and test-time adaptation of foundation models [8].

Reza Haffari is a Professor and previous ARC Future Fellow in the Department of Data Science & AI at Monash University (`Gholamreza.Haffari@monash.edu`). His research spans continual and lifelong learning [1], active experience selection and interaction, and uncertainty-aware self-evaluation and reliable adaptation [12].

References

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